

# JIOCX VS EXOTEL: COST & API INTEGRATION REPORT

Comparative Study for 1000 SMS/Month OTP Deliveries in India

|                    |                                  |                       |                                 |
|--------------------|----------------------------------|-----------------------|---------------------------------|
| Document Reference | LAW-SMS-COST-API-v1.1            | Target Platform       | SMS Gateway Integration (India) |
| Analysis Volume    | 1000 SMS / Month (Transactional) | Security / Regulatory | TRAI & DPDP Act Compliant       |

## 1. Executive Summary & Verdict

For a baseline volume of **1000 SMS/month** (primarily verification OTPs and notification alerts), both **Exotel** and **JioCX** (Jio Customer Experience) provide highly reliable cloud SMS gateways. However, their pricing structures, onboarding barriers, and developer support models are significantly different.

-  **Summary Recommendation:**
  - Use **Exotel** if you want a fast developer setup, pay-as-you-go credits without monthly commitments, and excellent API documentation.
  - Use **JioCX** if you already have corporate relationships with Reliance Jio, want a lower per-SMS rate for future high-volume scaling, or want free DLT registration.

## 2. Detailed Cost Analysis

Here is a breakdown of the expenses for sending **1000 SMS per month** under both gateways. Prices are in Indian Rupees (INR) and exclude government GST (18%).

| Cost Component           | Exotel SMS Gateway                     | JioCX SMS Gateway                         |
|--------------------------|----------------------------------------|-------------------------------------------|
| Average Cost per SMS     | ₹0.22 / SMS                            | ₹0.15 / SMS                               |
| DLT Registration Fee     | ₹5,900 (Paid to third-party registrar) | FREE (If registered directly via Jio DLT) |
| Monthly Cost (1,000 SMS) | ₹220 / month                           | ₹150 / month                              |
| Monthly Cost (5,000 SMS) | ₹1,100 / month                         | ₹750 / month                              |

| Cost Component             | Exotel SMS Gateway                  | JioCX SMS Gateway                                       |
|----------------------------|-------------------------------------|---------------------------------------------------------|
| Minimum Monthly Commitment | <b>None</b> (Pay-as-you-go credits) | May require minimum contract levels for small entities. |
| Initial Account Setup Time | 1 - 2 business days                 | 3 - 7 business days                                     |

### 3. Non-Technical Feature Comparison

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Beyond costs, you must consider the operational differences when running the service day-to-day.

#### 3.1. DLT Portal Alignment

In India, all SMS messages are checked against a Distributed Ledger Technology (DLT) database. Jio operates its own DLT portal (\*\*TrueConnect\*\*). If you use \*\*JioCX\*\*, your approval workflow for headers and templates is integrated within the same Jio ecosystem, reducing verification delays. With \*\*Exotel\*\*, you register on any DLT portal (such as Jio or PingConnect) and then copy-paste your IDs into Exotel's settings.

#### 3.2. Minimum Recharge & Account Maintenance

- **Exotel:** Allows buying SMS credits in small top-ups (minimum ₹500 to ₹1000). There are no recurring platform subscription fees for sending SMS (voice calling may require virtual number rentals, but SMS-only does not).
- **JioCX:** Primarily aimed at medium-to-large enterprises. Setup requires manual onboarding through a Jio sales representative, which can be slow for early-stage startups.

## 4. Technical Integration & APIs Needed

To implement the SMS OTP service, the system needs two backend endpoints and one external gateway call.

### 4.1. The User Flow (API Workflow)

1. **Step 1:** The user enters their phone number on the React app and clicks **"Send OTP"**.
2. **Step 2:** The React frontend makes an HTTP POST request to your backend: `/api/auth/send-sms-otp`.
3. **Step 3:** The backend generates a 6-digit code, saves it with a 5-minute expiry in the database, and calls either the Exotel or JioCX API.
4. **Step 4:** The user receives the SMS, enters the OTP, and clicks **"Verify"**. The frontend calls: `/api/auth/verify-sms-otp`.

### 4.2. Option A: Exotel API Specification

The developer uses Basic Authentication with your Exotel API Key and Token to send the message payload:

```
// POST request to Exotel SMS Gateway
URL: https://api.exotel.com/v1/Accounts/{your_account_sid}/Sms/send.json
Method: POST
Headers:
  Authorization: Basic {Base64_Encoded(API_Key:API-Token)}
  Content-Type: application/x-www-form-urlencoded

Payload Parameters:
  From: LWMATE (Your approved 6-character DLT Header)
  To: +919876543210 (User phone number)
  Body: 123456 is your OTP for LawOnCall. Valid for 5 mins.
  DltEntityId: 1402xxxxxx (Your 15-digit DLT Entity ID)
  DltTemplateId: 1207xxxxxx (Your approved DLT Template ID)
```

### 4.3. Option B: JioCX API Specification

JioCX uses a REST API with Bearer token authentication or custom header credentials:

```
// POST request to JioCX SMS Gateway
URL: https://jiocx.reliancejio.com/api/v2/sms/send
Method: POST
Headers:
  Authorization: Bearer {JioCX_Access_Token}
  Content-Type: application/json

Payload (JSON format):
{
  "senderId": "LWMATE",
  "templateId": "1207xxxxxxxxxxxxxx",
  "entityId": "1402xxxxxxxxxxxxxx",
  "destination": "+919876543210",
  "message": "123456 is your OTP for LawOnCall. Valid for 5 mins."
}
```

## 5. Compliance & Security (TRAI & DPDP Act 2023)

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Since the application processes user phone numbers and delivers sensitive verification OTPs, compliance with Indian telecom and privacy laws is mandatory:

- TRAI TCCCPR 2018: Requires strict scrub-matching of all message content against registered DLT templates. Both Exotel and JioCX automatically validate these templates at the carrier gateway layer.
- Digital Personal Data Protection (DPDP) Act, 2023: Phone numbers constitute Personal Identifiable Information (PII). Both Exotel and JioCX encrypt data in transit (TLS 1.2/1.3) and support strict access-control logs to prevent unauthorized data exposure.